



Round 1: “Pop Culture”

Advanced Division

Name: _____

A.1.1	Sudoku	15 points	A.1.10	TomTom	20 points
A.1.2	Sudoku	80 points	A.1.11	TomTom	60 points
A.1.3	Sudoku	180 points	A.1.12	TomTom	200 points
A.1.4	Numberlink	20 points	A.1.13	Battleships	10 points
A.1.5	Numberlink	55 points	A.1.14	Battleships	30 points
A.1.6	Numberlink	70 points	A.1.15	Battleships	50 points
A.1.7	Domino Hunt	45 points			
A.1.8	Domino Hunt	65 points			
A.1.9	Domino Hunt	100 points			

Total: 1000 points

A.1.1 – A.1.3: Sudoku

Insert a number from 1 to 9 (1 to 6 in the first puzzle) so that no number repeats in any row, column, or box.

Puzzle A.1.1 (15 points)

	4	1	3		
	3			6	
	5			3	
		3	1	2	

Puzzle A.1.2 (80 points)

					1			
		9	8		2	3		
	7	1				4	5	
8	4						6	
	6						4	1
	2	3				5	8	
		8	6		3	7		
			4					

Puzzle A.1.3 (180 points)

5				8				9
			2		7		8	
		4				7		
	7				6			
9				5				6
			4				3	
		3				6		
	2		8		9			
1				6				3

A.1.4 – A.1.6: Numberlink

Draw non-intersecting paths through the centers of some cells, each connecting one clue to its equal counterpart.

(Note: all cells will be used in the final solutions to these puzzles.)

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A.1.4 – A.1.6: Numberlink

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[illegible]

	1						2	3
			4	5	3			
						6	4	
		2	1					
				7	8	6		
	7	8						5

[illegible]

A.1.7 – A.1.9: Domino Hunt

Divide the grid into dominoes so that each domino in the indicated set is used exactly as many times as it appears. The orientation of the digits does not matter.

Puzzle A.1.7 (45 points)

1	1	3	1	3	1
3	3	2	3	3	1
3	1	2	2	2	3
1	3	2	2	2	2
1	3	2	1	2	3
1	1	1	3	2	2

1 1	1 1	1 1
1 2	1 2	1 2
1 3	1 3	1 3
2 2	2 2	2 2
2 3	2 3	2 3
3 3	3 3	3 3

Puzzle A.1.8 (65 points)

0	1	3	5	0	5	3	3
2	6	6	0	0	2	1	0
2	6	3	1	4	1	4	4
6	5	4	4	4	1	4	3
3	3	2	2	5	5	5	1
0	1	2	6	4	5	6	1
3	6	6	0	5	2	2	0

0 0						
0 1	1 1					
0 2	1 2	2 2				
0 3	1 3	2 3	3 3			
0 4	1 4	2 4	3 4	4 4		
0 5	1 5	2 5	3 5	4 5	5 5	
0 6	1 6	2 6	3 6	4 6	5 6	6 6

Puzzle A.1.9 (100 points)

3	6	7	8	1	4	5	8
2	5	4	7	6	3	2	7
7	4	1	2	5	8	3	6
6	8	2	5	4	6	3	2
7	1	3	7	8	7	4	7
2	5	4	6	3	5	2	1
3	6	1	8	5	6	3	8
4	1	2	1	8	1	4	5

1 2	3 2	5 2	7 2
1 2	3 2	5 2	7 2
1 4	3 4	5 4	7 4
1 4	3 4	5 4	7 4
1 6	3 6	5 6	7 6
1 6	3 6	5 6	7 6
1 8	3 8	5 8	7 8
1 8	3 8	5 8	7 8

A.1.10 - A.1.12: TomTom

Place a number from 1 to N into each cell so that each row and column contains every number from that range with no repeats, where N is the side length of the grid. A clue represents the value obtained by applying an operation iteratively on the numbers in the region the clue is in. If no operation is given, it may be any of $+$, $-$, $\times$, or $\div$. Subtraction and division in regions with more than two numbers are handled by taking the largest number and subtracting/dividing all the others.

Puzzle A.1.10 (20 points)

8×		6×	8+	
	9+			
		6+		8×
8+				

Puzzle A.1.11 (60 points)

8				8
8				
8		8		
		8		

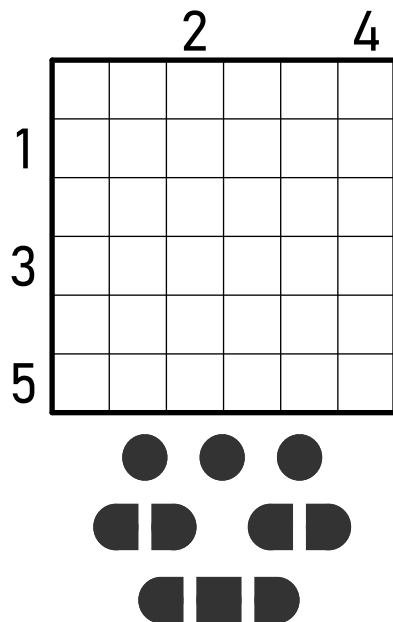
Puzzle A.1.12 (200 points)

	20		26		
					2
20	26		0		
				2	
					6

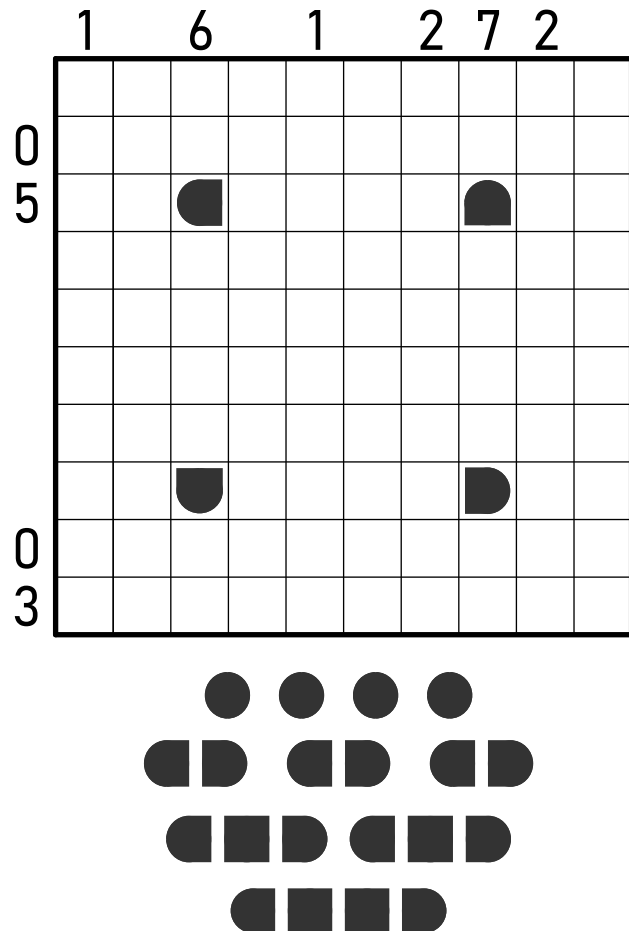
A.1.13 - A.1.15: Battleships

Place the given fleet of ships into the grid so that no two ships are touching, not even diagonally. Rotating ships is permitted. A clue outside the grid indicates the number of cells in the corresponding row or column that are occupied by ships. Cells with waves cannot be occupied by a ship. A given ship segment must be used as the part of a ship that its shape represents.

Puzzle A.1.13 (10 points)



Puzzle A.1.14 (30 points)



Puzzle A.1.15 (50 points)

